

```
DROP DATABASE IF EXISTS SouthernOhioInfoWarehouse;
```

```
CREATE DATABASE SouthernOhioInfoWarehouse;
```

```
USE SouthernOhioInfoWarehouse;
```

```
CREATE TABLE Facility(  
    FacilityID INT AUTO_INCREMENT PRIMARY KEY,  
    FacilityName VARCHAR(50),  
    County VARCHAR (30),  
    City VARCHAR (30),  
    State CHAR (2),  
    StreetAddress VARCHAR(50)  
);
```

```
CREATE TABLE Equipment(  
    EquipmentID INT AUTO_INCREMENT PRIMARY KEY,  
    EquipmentName VARCHAR(40),  
    EquipmentType VARCHAR(40),  
    PurchaseDate DATE,  
    EquipmentStatus VARCHAR(40),  
    EquipmentSerialNumber VARCHAR(40),  
    ModelNumber VARCHAR(40),  
    FacilityID INT,  
    CONSTRAINT FK_Equipment_Facility FOREIGN KEY (FacilityID) REFERENCES Facility  
    (FacilityID)
```

);

```
CREATE TABLE InspectionRecord(  
    InspectionID INT AUTO_INCREMENT PRIMARY KEY,  
    InspectionDate DATE,  
    InspectorFirstName VARCHAR(30),  
    InspectorLastName VARCHAR(30),  
    InspectionType VARCHAR(30),  
    Result CHAR(1),  
    EquipmentID INT,  
    CONSTRAINT FK_InspectionRecord_Equipment FOREIGN KEY (EquipmentID)  
    REFERENCES Equipment (EquipmentID)  
);
```

```
CREATE TABLE Employee(  
    EmployeeID INT AUTO_INCREMENT PRIMARY KEY,  
    EmployeeFirstName VARCHAR(30),  
    EmployeeLastName VARCHAR(30),  
    PhoneNumber VARCHAR(20),  
    StreetAddress VARCHAR(50),  
    EmailAddress VARCHAR(100),  
    JobTitle VARCHAR(50),  
    Department VARCHAR(50),  
    HireDate DATE,  
    FacilityID INT,
```

```
CONSTRAINT FK_Employee_Facility FOREIGN KEY (FacilityID) REFERENCES Facility
(FacilityID)

);
```

```
CREATE TABLE MaintenanceRecord(
MaintenanceID INT AUTO_INCREMENT PRIMARY KEY,
MaintenanceDate DATE,
TechnicianFirstName VARCHAR(30),
TechnicianLastName VARCHAR(30),
Description VARCHAR(250),
Cost DECIMAL(10, 2),
WorkOrderNumber VARCHAR(30),
EquipmentID INT,
MaintenanceStatus VARCHAR(20),
CONSTRAINT FK_MaintenanceRecord_Equipment FOREIGN KEY (EquipmentID)
REFERENCES Equipment (EquipmentID)
);
```

```
INSERT INTO FACILITY
(FacilityName, County, City, State, StreetAddress)
VALUES
('Piketon Operations Center', 'Pike', 'Piketon', 'OH', '100 Energy Drive'),
('Scioto Maintenance Hub', 'Scioto', 'Portsmouth', 'OH', '250 Industrial Blvd'),
('Ross Regional Service Center', 'Ross', 'Chillicothe', 'OH', '75 Commerce Way');
```

```
SELECT * FROM Facility;
```

INSERT INTO EQUIPMENT

(EquipmentID, FacilityID, EquipmentName)

VALUES

(1, 1, 'Transformer T-101'),

(2, 1, 'Cooling Pump CP-12'),

(3, 2, 'Conveyor Motor M-8'),

(4, 3, 'Backup Generator');

SELECT * FROM EQUIPMENT;

INSERT INTO EMPLOYEE

(EmployeeID, FacilityID, EmployeeFirstName, EmployeeLastName, Department)

VALUES

(1, 1, 'Sarah', 'Jones', 'Maintenance'),

(2, 1, 'Mike', 'Davis', 'Operations'),

(3, 2, 'Emily', 'Brown', 'Safety');

SELECT * FROM EMPLOYEE;

INSERT INTO InspectionRecord

(InspectionDate, InspectorFirstName, InspectorLastName, InspectionType, Result, EquipmentID)

VALUES

('2026-06-01', 'Emily', 'Brown', 'Safety', 'P', 2),

('2026-06-05', 'Sarah', 'Jones', 'Mechanical', 'F', 3),

```
('2026-06-08', 'Mike', 'Davis', 'Routine', 'P', 4);
```

```
SELECT * FROM INSPECTIONRECORD;
```

```
INSERT INTO MAINTENANCERECORD
```

```
(MaintenanceDate, TechnicianFirstName, TechnicianLastName, Description, Cost,  
WorkOrderNumber, EquipmentID, MaintenanceStatus)
```

```
VALUES
```

```
('2026-06-10', 'Sarah', 'Jones', 'Oil Change', 125.00, 'WO-1001', 1, 'Completed'),
```

```
('2026-06-12', 'Mike', 'Davis', 'Bearing replacement', 840.50, 'WO-1002', 3, 'Completed'),
```

```
('2026-06-14', 'Emily', 'Brown', 'Routine Lubrication', 75.00, 'WO-1003', 2, 'Completed');
```

```
SELECT * FROM MAINTENANCERECORD;
```

1. Find all inspections failed inspections, the right answer

```
SELECT
```

```
    ir.EquipmentID,
```

```
    e.EquipmentName,
```

```
    ir.InspectionDate,
```

```
    ir.Result,
```

```
    ir.InspectionType
```

```
FROM inspectionrecord AS ir
```

```
JOIN equipment AS e
```

```
ON ir.EquipmentID = e.EquipmentID
```

```
WHERE ir.Result = 'F';
```

Things to remember for #1 that I forgot when I typed it

- Remember to put the table names and a dot before the attributes as you select them
- Remember to get the equipment name

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	EquipmentID	EquipmentName	InspectionDate	Result	InspectionType
▶	3	Conveyor Motor M-8	2026-06-05	F	Mechanical

Facility 1

EQUIPMENT 2

EMPLOYEE 3

INSPECTIONRECORD 4

Result 5 ×

Res

2. Show all inspection records, the right answer

SELECT

e.EquipmentID,
e.EquipmentName,
ir.InspectionID,
ir.InspectionDate,
ir.InspectorFirstName,
ir.InspectorLastName,
ir.InspectionType,
ir.Result

FROM inspectionrecord AS ir

JOIN equipment AS e

ON ir.EquipmentID = e.EquipmentID;

Things to remember from query #2 that I forgot when I typed it

- Check the create table statement before writing the select list
- Use the exact column names that exist in the table
- InspectionREcord does not have one Inspector column, it has inspector firstname and inspector last name
- Use commas between all selected columns except the last one

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	EquipmentID	EquipmentName	InspectionID	InspectionDate	InspectorFirstName	InspectorLastName	InspectionType	Result
▶	2	Cooling Pump CP-12	1	2026-06-01	Emily	Brown	Safety	P
	3	Conveyor Motor M-8	2	2026-06-05	Sarah	Jones	Mechanical	F
	4	Backup Generator	3	2026-06-08	Mike	Davis	Routine	P

3.Which single maintenance will cost the most? The right answer

SELECT

e.EquipmentName,

mr.Description,

mr.Cost

FROM maintenancerecord AS mr

JOIN equipment AS e

ON mr.EquipmentID = e.EquipmentID

ORDER BY mr.Cost DESC

LIMIT 1;

Things to remember for query #3

- Use the ERD to identify which table owns each attribute
- Cost is in MaintenanceRecord, not Equipment
- EquipmentName is in Equipment
- Table names cannot have spaces unless created with spaces
- MaintenanceRecord connects to Equipment through EquipmentID
- To find the highest values and OrderBy Desc Limit 1
- If asking for total cost per equipment, use SUM() and GROUP BY

Result Grid			
	EquipmentName	Description	Cost
▶	Conveyor Motor M-8	Bearing replacement	840.50

4. how much has each facility spent on maintenance, my right answer

SELECT

f.FacilityID,

f.FacilityName,

SUM(mr.Cost) AS TotalMaintenanceCost

FROM facility AS f

JOIN equipment AS e

ON f.FacilityID = e.FacilityID

JOIN maintenancerecord AS mr

ON e.EquipmentID = mr.EquipmentID

GROUP BY

f.FacilityID,

f.FacilityName

ORDER BY TotalMaintenanceCost DESC;

Things to remember for Query #4:

- The mistake was using a 2nd from, sql uses 1 from for multiple joins
- Because the question says “each facility” use GROUP BY facility
- Because the question asks how much, the question asks “how much” use SUM()
- Watch spelling: Facility not facility
- Don't put 2 commas after FacilityName,

5. Which equipment is located at each facility, the correct answer

SELECT

 e.EquipmentID,
 e.EquipmentName,
 f.FacilityID,
 f.FacilityName

FROM equipment AS e

JOIN facility AS f

ON e.FacilityID = f.FacilityID

ORDER BY

 f.FacilityName,
 e.EquipmentName;

Things to remember for query #5

- I chose the right tables: Equipment and Facility
- I chose the right join: Equipment.FacilityID = Facility.FacilityID
- If I use aliases like e and f, I must define them in FROM/JOIN
- Do not put a comma after the last selected column
- This one was mostly a syntax issue not a logic issue

Result Grid			
Filter Rows:			
Export:			
	FacilityID	FacilityName	TotalMaintenanceCost
▶	2	Scioto Maintenance Hub	840.50
	1	Piketon Operations Center	200.00

6. Which employees work at each facility: the right answer

SELECT

e.EmployeeID,

e.EmployeeFirstName,

e.EmployeeLastName,

e.Department,

f.FacilityID,

f.FacilityName

FROM facility AS f

JOIN employee AS e

ON f.FacilityID = e.FacilityID

ORDER BY

f.FacilityName,

e.EmployeeLastName;

Key lessons for Query #6

- Use EmployeeFirstName and EmployeeLastName, not FirstName and LastName
- Do not use group by unless you are summarizing the data with COUNT, SUM, AVG, etc.
- Remove the comma after the last selected column
- You were close, the main issue was mainly the column-name precisoin and knowing when GROUP BY is needed

Result Grid Filter Rows: Export: Wrap Cell Content:						
	EmployeeID	EmployeeFirstName	EmployeeLastName	Department	FacilityID	FacilityName
▶	2	Mike	Davis	Operations	1	Piketon Operations Center
	1	Sarah	Jones	Maintenance	1	Piketon Operations Center
	3	Emily	Brown	Safety	2	Scioto Maintenance Hub

7. Which equipment has never failed inspection, the right answer

SELECT

e.EquipmentID,

e.EquipmentName

FROM equipment AS e

WHERE e.EquipmentID NOT IN

(

SELECT

ir.EquipmentID

FROM inspectionrecord AS ir

WHERE ir.Result = 'F'

);

What I did wrong for Question 7

- Show me equipment where the ID is not in the list of equipment IDs that failed inspection
- I used Result != 'F' which finds passed, records, not equipment that never failed
- You had alias r in FROM/JOIN but then used ir in SELECT/ON
- You u had a comma before teh last select column
- Never failed is different from “this inspection passed”

- Use NOT IN or NOT EXISTS to exclude equipment that appears in failed inspection records
- Be consistent with aliases: if I say InspectionRecord AS ir, then use ir everywhere.
- Do not put a comma after the last selected column

Result Grid		Filter Rows:
	EquipmentID	EquipmentName
▶	1	Transformer T-101
	2	Cooling Pump CP-12
	4	Backup Generator
✱	NULL	NULL

8. Which inspections occurred in June, the right answer

SELECT

ir.InspectionID,

ir.InspectionDate,

e.EquipmentID,

e.EquipmentName

FROM inspectionrecord AS ir

JOIN equipment AS e

ON ir.EquipmentID = e.EquipmentID

WHERE ir.InspectionDate BETWEEN '2026-06-01' AND '2026-06-30';

What I did wrong in question 8

- Use different aliases for different tables
- If Inspection record as ir, then equipment should be AS e,
- SQL dates should use YYYY-MM-DD format
- Use BETWEEN when filtering for a date range
- Do not put a comma after the last select range

